WEEK 2 · MODULE 1 — FOUNDATIONS

Prompting, Structured Output & Tool Calling

Bridge · first code *You start writing code this week*

Goal: Build a small app that talks to an AI, gets tidy answers back, and can do one real action.

What this week is about

Now you write your first code. You'll learn how to talk to the AI properly — giving it clear instructions, getting answers back in a tidy format your program can use, and letting it do one real action like a lookup.

By Friday you'll have a small working app.

Why it matters

Isn't “prompting” just typing a question?

For a toy, yes. For a product, no — a good prompt is written carefully, with examples, so the AI behaves the same way every time.

Why do I need answers in a fixed format?

If the AI replies in free text, your program can't reliably read it. A fixed format (like JSON) lets your code actually use the answer.

What is “tool calling”?

It lets the AI ask your code to do something — look up an order, do a calculation — and use the result. The AI asks; your code does the work.

What you'll learn

- **Writing a clear prompt** — and why good examples beat long instructions.
- **Your first real call to the AI from code** — keys kept safe, never in your files.
- **Fixed-format answers (JSON)** your program can trust.
- **Validation and retry:** checking the answer is valid, and asking again politely when it isn't.
- **Tool calling:** letting the AI trigger a real action — and knowing your code, not the AI, runs it.
- **Simple guardrails:** catching bad input, and refusing safely instead of making things up.

Topics covered

The exact concepts to study this week:

Prompt anatomy	Zero-shot vs few-shot	Chain-of-Thought (CoT)
Self-consistency	Task decomposition	Structured output (JSON schema)
Pydantic	instructor library	Validation & retry
Tool / function calling	Parallel tool calls	Guardrails
Prompt injection		

How your mentor checks

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FRIDAY · NO MARKS

A quick review of your task. Not a test — just a few things your mentor looks at, so you can fix anything before Monday.

- ☐ Does the app return answers in a clean, fixed format every time?
- ☐ When given a broken input, does it fail safely instead of crashing or inventing?
- ☐ Can you explain, from your own code, who actually runs the tool — the AI or your code?
- ☐ Is the API key kept out of your code files?